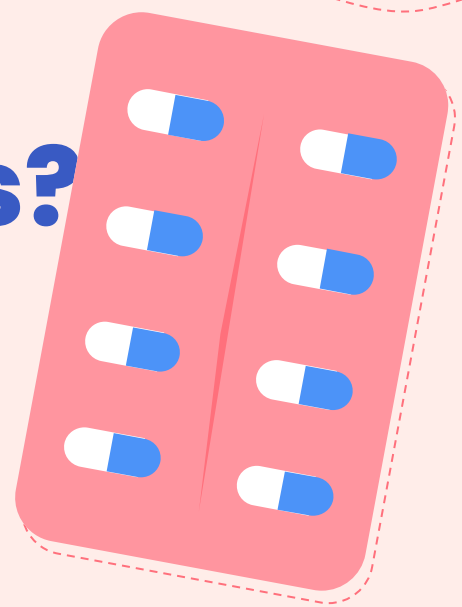
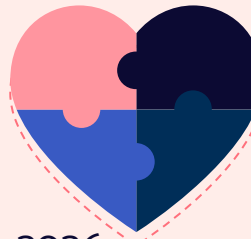
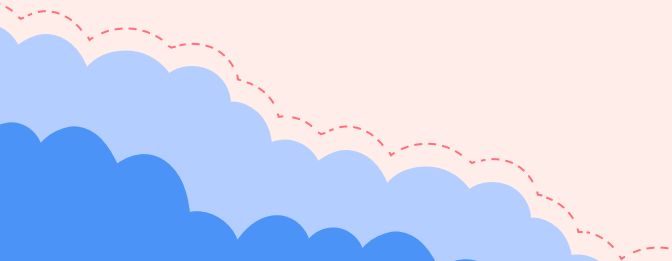
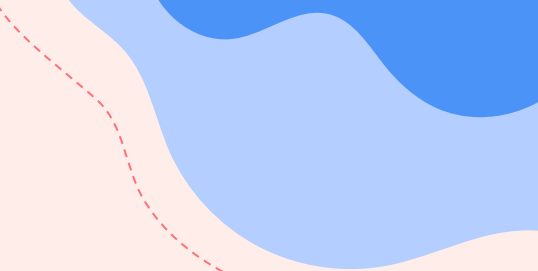




Can we predict depression in students?

A machine learning approach

Sneha Sivakaran | Module 12 - UMIT Tirol | 12th June 2026



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Research Question & Dataset



Research Question

Can lifestyle and academic factors such as sleep duration, academic pressure and financial stress be used to predict depression in university students?



Dataset

Source: Kaggle- Student Depression Dataset

Entries: 503

Features: 11

Target: Depression (Yes / No)

Methods: Logistic Regression & Random Forest

Preprocessing & EDA

```
# Check for missing values
print("Missing values:")
print(df.isnull().sum())
```



```
Missing values:
Gender          0
Age             0
Academic Pressure  0
Study Satisfaction  0
Sleep Duration   0
Dietary Habits    0
Have you ever had suicidal thoughts ?  0
Study Hours       0
Financial Stress  0
Family History of Mental Illness  0
Depression        0
dtype: int64
```

```
# Check 0-values for biological plausibility (as required!)
# Sleep Duration = 0 would be biologically impossible -> check
print("Zero values per column:")
print[(df == 0).sum()]
```



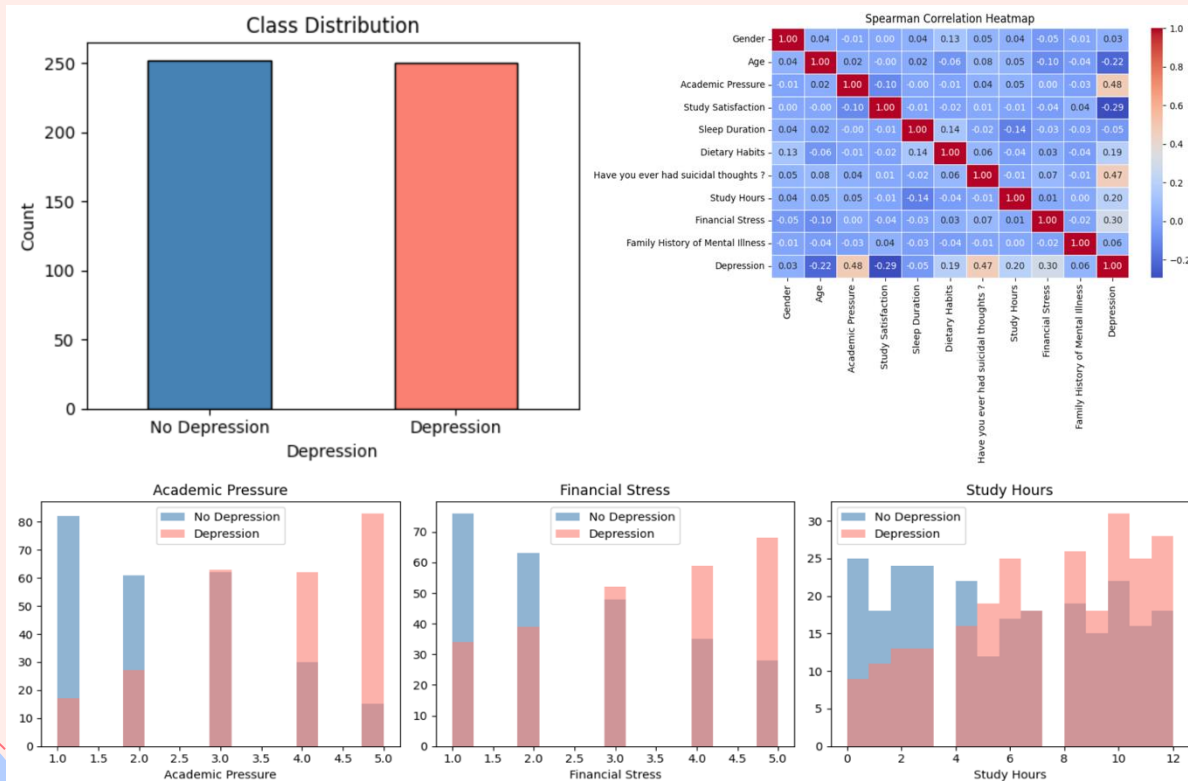
```
Zero values per column:
Gender          0
Age             0
Academic Pressure  0
Study Satisfaction  0
Sleep Duration    0
Dietary Habits     0
Have you ever had suicidal thoughts ?  0
Study Hours       34
Financial Stress   0
Family History of Mental Illness  0
Depression         0
dtype: int64
```

```
# Check normality with Shapiro-Wilk test -> decides Spearman vs Pearson
numeric_cols = df_encoded.select_dtypes(include=np.number).columns.tolist()
print("Shapiro-Wilk Normality Test:")
for col in numeric_cols:
    stat, p = stats.shapiro(df_encoded[col])
    normal = "normal" if p > 0.05 else "NOT normal"
    print(f" {col}: p={p:.4f} -> {normal}")
```



```
Shapiro-Wilk Normality Test:
Gender: p=0.0000 -> NOT normal
Age: p=0.0000 -> NOT normal
Academic Pressure: p=0.0000 -> NOT normal
Study Satisfaction: p=0.0000 -> NOT normal
Sleep Duration: p=0.0000 -> NOT normal
Dietary Habits: p=0.0000 -> NOT normal
Have you ever had suicidal thoughts ?: p=0.0000 -> NOT normal
Study Hours: p=0.0000 -> NOT normal
Financial Stress: p=0.0000 -> NOT normal
Family History of Mental Illness: p=0.0000 -> NOT normal
Depression: p=0.0000 -> NOT normal
```

Preprocessing & EDA



Modelling & Evaluation

```
# Logistic Regression with class_weight='balanced' to handle class imbalance
lr = LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42)
lr.fit(X_train_scaled, y_train)

y_pred_lr = lr.predict(X_test_scaled)
y_prob_lr = lr.predict_proba(X_test_scaled)[:, 1]

print("=== Logistic Regression ===")
print(classification_report(y_test, y_pred_lr, target_names=['No Depression', 'Depression']))
print("ROC-AUC:", round(roc_auc_score(y_test, y_prob_lr), 4))
```



```
=== Logistic Regression ===
```

	precision	recall	f1-score	support
No Depression	0.97	1.00	0.98	63
Depression	1.00	0.97	0.98	63
accuracy			0.98	126
macro avg	0.98	0.98	0.98	126
weighted avg	0.98	0.98	0.98	126

ROC-AUC: 0.9965

```
# Random Forest with class_weight='balanced' to handle class imbalance
rf = RandomForestClassifier(n_estimators=100, class_weight='balanced', random_state=42)
rf.fit(X_train_scaled, y_train)

y_pred_rf = rf.predict(X_test_scaled)
y_prob_rf = rf.predict_proba(X_test_scaled)[:, 1]

print("=== Random Forest ===")
print(classification_report(y_test, y_pred_rf, target_names=['No Depression', 'Depression']))
print("ROC-AUC:", round(roc_auc_score(y_test, y_prob_rf), 4))
```



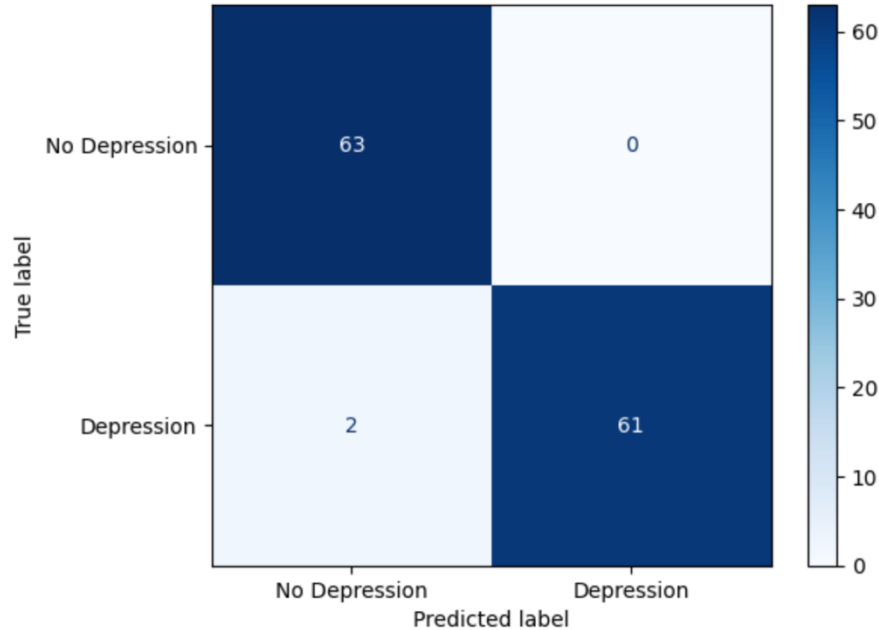
```
=== Random Forest ===
```

	precision	recall	f1-score	support
No Depression	0.98	0.89	0.93	63
Depression	0.90	0.98	0.94	63
accuracy			0.94	126
macro avg	0.94	0.94	0.94	126
weighted avg	0.94	0.94	0.94	126

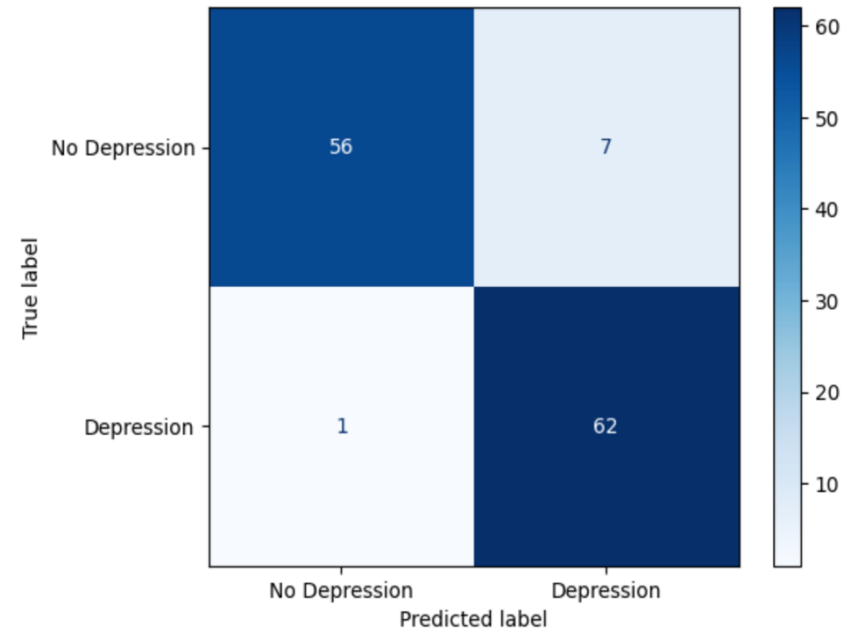
ROC-AUC: 0.9864

Modelling & Evaluation

Confusion Matrix - Logistic Regression

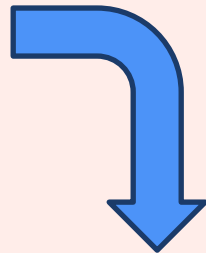


Confusion Matrix - Random Forest



Model Comparison

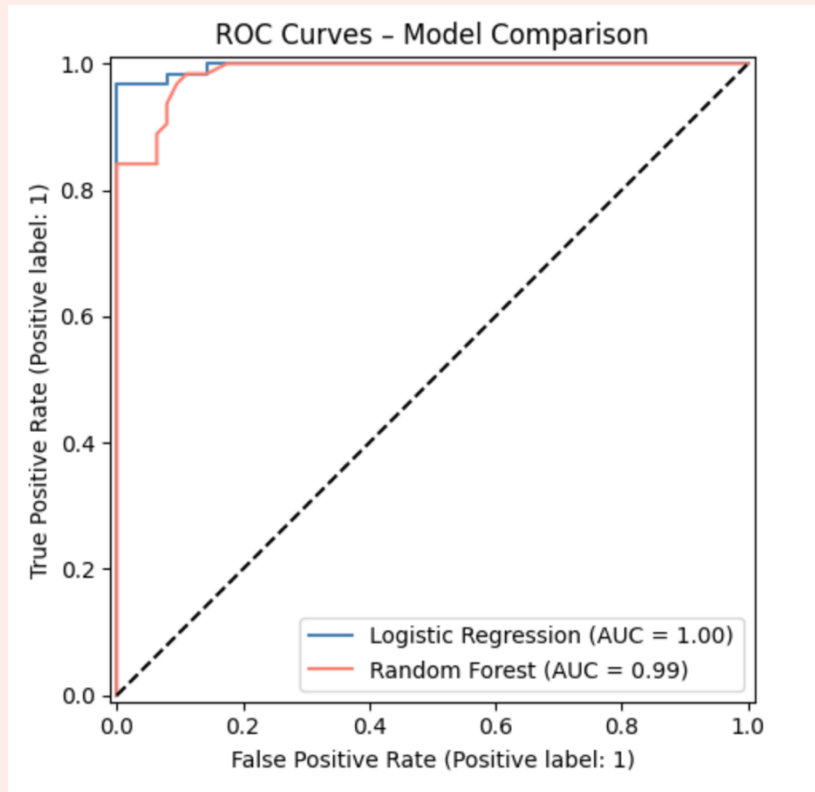
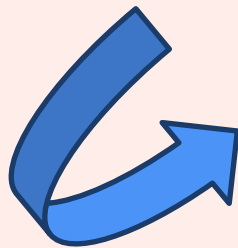
```
# Compare both models side by side
results = pd.DataFrame({
    'Model': ['Logistic Regression', 'Random Forest'],
    'F1-Score': [round(f1_score(y_test, y_pred_lr), 4),
                 round(f1_score(y_test, y_pred_rf), 4)],
    'ROC-AUC': [round(roc_auc_score(y_test, y_prob_lr), 4),
                round(roc_auc_score(y_test, y_prob_rf), 4)]
})
print(results.to_string(index=False))
```



Model	F1-Score	ROC-AUC
Logistic Regression	0.9839	0.9965
Random Forest	0.9394	0.9864

ROC Curves

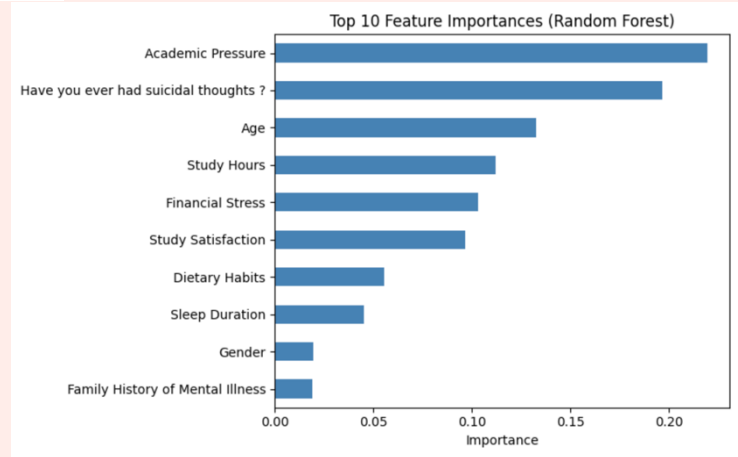
```
# ROC Curves for both models in one plot
fig, ax = plt.subplots(figsize=(7, 5))
RocCurveDisplay.from_estimator(lr, X_test_scaled, y_test,
                              name='Logistic Regression', ax=ax, color='steelblue')
RocCurveDisplay.from_estimator(rf, X_test_scaled, y_test,
                              name='Random Forest', ax=ax, color='salmon')
ax.plot([0, 1], [0, 1], 'k--', label='Random Chance')
plt.title("ROC Curves - Model Comparison")
plt.tight_layout()
plt.show()
```



Feature Importance

```
# Feature Importance – which factors predict depression most?
importances = pd.Series(rf.feature_importances_, index=X.columns)
top10 = importances.nlargest(10)

plt.figure(figsize=(8, 5))
top10.sort_values().plot(kind='barh', color='steelblue')
plt.xlabel("Importance")
plt.title("Top 10 Feature Importances (Random Forest)")
plt.tight_layout()
plt.show()
```



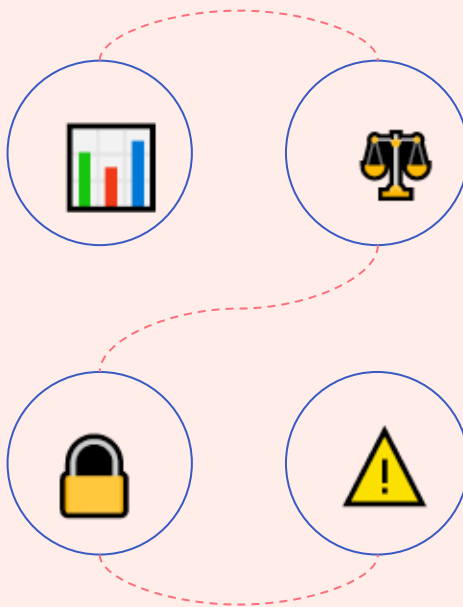
Ethics & Bias

Data Bias

- Only university students
- Not generalisable to other populations
- (working adults, different cultures, school children)

Sensitive Data

- Mental health = highly sensitive data
- GDPR compliance required
- Informed consent mandatory



Algorithmic Bias

- Underrepresented subgroups (age, gender)
- Model may perform worse for minority groups
- No demographic fairness testing done

False Predictions

- **False Negative** → depressed student goes undetected
- **False Positive** → student flagged incorrectly, wastes resources

EU AI Act

How does this system fit in?

High- Risk AI System

- Mental health prediction
- Classified under EU AI Act

High-Risk Category

- Annex III: health & education
- Strict requirements apply

Transparency

- Decisions must be explainable
- Results must be documented

Human Oversight

- Counsellor/doctor reviews output
- No fully automated decisions

No Autonomous Decisions

- Screening tool only
- Never sole basis for diagnosis

Accountability

- Developer & deployer responsible
- Legal liability applies

Conclusion

"Can lifestyle and academic factors predict depression in university students?"

→ Yes, they can!

- Academic Pressure, Financial Stress & Sleep
- Duration are the **strongest predictors**
- Logistic Regression: **98% accuracy, ROC-AUC 0.9965** → best overall model
- Random Forest: slightly lower but strong, only **1 False Negative**
- Early ML-based detection could enable **targeted student support programmes**
- Ethical deployment requires human oversight & data protection



Resources

- Eissler, C., Sailer, M., Walter, S. & Jerg-Bretzke, L. (2019). Psychische Gesundheit und Belastung bei Studierenden. Prävention und Gesundheitsförderung, 15(3), 242–249.
<https://doi.org/10.1007/s11553-019-00746-z>
- Mofatteh, M. (2020). Risk factors associated with stress, anxiety, and depression among university undergraduate students. AIMS Public Health, 8(1), 36–65.
<https://doi.org/10.3934/publichealth.2021004>
- Weber, R. (2024, 26. Februar). Wie belastet sind Studierende? Forschung & Lehre.
<https://www.forschung-und-lehre.de/zeitfragen/wie-belastet-sind-studierende-6268>



**Thank you
for your
attention!**

